

$$\int_0^1 \frac{\sin^{-1} x \cos^{-1} x}{\sin^{-1} x + \cos^{-1} x} dx$$

$$\frac{2}{\pi} \int_0^1 \sin^{-1} x \left(\frac{\pi}{2} - \sin^{-1} x \right) dx$$

$$\frac{2}{\pi} \int_0^1 \left(\frac{\pi}{2} \arcsin x - \arcsin^2 x \right) dx$$